RACE Regular Analysis Through Continuous Exercise

Current Electricity : RACE-01

1. The current in a conductor is expressed as $I = 3t^2 + 4t^3$, where I is in Ampere and t is in second. The amount of electric charge that flows through a section of the conductor during $t = 1s$ to $t = 2s$ is _____ C.

PCT149

2. The electric current through a wire varies with time as $I = I_0 + \beta t$. where $I_0 = 20\ A$ and $\beta = 3\ A/s$. The amount of electric charge crossed through a section of the wire in 20 s is :

(A) 80 C (B) 1000 C (C) 800 C (D) 1600 C

PCT150

3. When $5V$ potential difference is applied across a wire of length 0.1 m, the drift speed of electrons is $2.5 \times 10^{-4}\ ms^{-1}$. If the electron density in the wire is $8 \times 10^{28}\ m^{-3}$, the resistivity of the material is close to :-

(A) $1.6 \times 10^{-6}\ \Omega m$ (B) $1.6 \times 10^{-5}\ \Omega m$ (C) $1.6 \times 10^{-8}\ \Omega m$ (D) $1.6 \times 10^{-7}\ \Omega m$

PCT151

4. Drift speed of electrons, when 1.5 A of current flows in a copper wire of cross section 5 mm^2, is v. If the electron density in copper is $9 \times 10^{28}/m^3$ the value of v in mm/s is close to

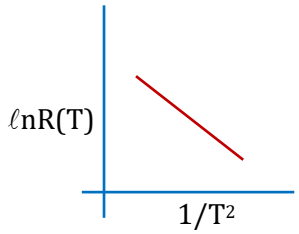
(Take charge of electron to be = $1.6 \times 10^{-19} C$)

(A) 0.2 (B) 3 (C) 2 (D) 0.02

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PCT152

5. In an experiment, the resistance of a material is plotted as a function of temperature (in some range). As shown in the figure, it is a straight line. One may conclude that :



(A) $R(T) = \frac{R_0}{T^2}$ (B) $R(T) = R_0 e^{-T^2/T_0^2}$ (C) $R(T) = R_0 e^{-T_0^2/T^2}$ (D) $R(T) = R_0 e^{T^2/T_0^2}$

PCT153

6. A cylindrical wire of radius 0.5 mm and conductivity $5 \times 10^7\ S/m$ is subjected to an electric field of $10\ mV/m$. The expected value of current in the wire will be $x^3\pi\ mA$. The value of x is ___.

PCT154

7. A conducting wire of length 'l', area of cross-section A and electric resistivity ρ is connected between the terminals of a battery. A potential difference V is developed between its ends, causing an electric current.

If the length of the wire of the same material is doubled and the area of cross-section is halved, the resultant current would be :

(A) $\frac{1}{4}\frac{VA}{\rho l}$ (B) $\frac{3}{4}\frac{VA}{\rho l}$ (C) $\frac{1}{4}\frac{\rho l}{VA}$ (D) $4\frac{VA}{\rho l}$

PCT155

For More Join: @IITJEE_Advanced

8. A current of 5 A is passing through a non-linear magnesium wire of cross-section 0.04 m^2. At every point the direction of current density is at an angle of 60° with the unit vector of area of cross-section. The magnitude of electric field at every point of the conductor is : (Resistivity of magnesium $\rho = 44 \times 10^{-8}\ \Omega m$)

(A) $11 \times 10^{-2}\ V/m$ (B) $11 \times 10^{-7}\ V/m$ (C) $11 \times 10^{-5}\ V/m$ (D) $11 \times 10^{-3}\ V/m$

PCT156

9. A. The drift velocity of electrons decreases with the increase in the temperature of conductor.

B. The drift velocity is inversely proportional to the area of cross-section of given conductor.

C. The drift velocity does not depend on the applied potential difference to the conductor.

D. The drift velocity of electron is inversely proportional to the length of the conductor.

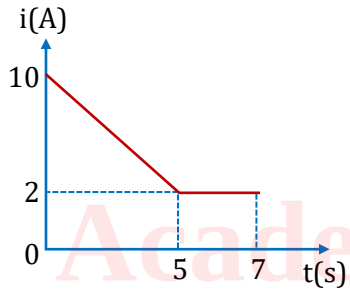
E. The drift velocity increases with the increase in the temperature of conductor.

Choose the correct answer from the options given below:

(A) A and B only (B) A and D only (C) B and E only (D) B and C only

PCT157

10. Current flowing in a wire varies with time as shown in the graph. Total charge flown through cross-section of the wire is $17n$. Find the value of n



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PCT158

11. A 150 m long metal wire connects points A and B. The electric potential at point B is 50 V less than that at point A. If the conductivity of the metal is $60 \times 10^6\ mho/m$, then magnitude of the current density in the wire is equal to $n \times 10^6\ A/m^2$

PCT159

ANSWER KEY

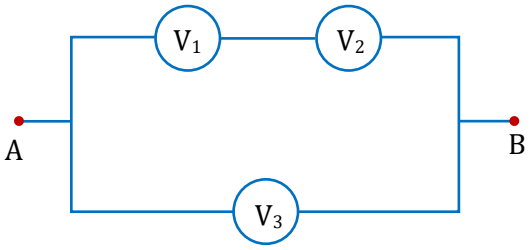
Question	1	2	3	4	5	6	7	8	9	10	11
Answer	22	B	B	D	C	5	A	C	B	2	20

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Current Electricity : RACE-02

1. Three voltmeters, all having different internal resistances are joined as shown in figure. When some potential difference is applied across A and B, their readings are V_1, V_2 and V_3. Choose the correct option.



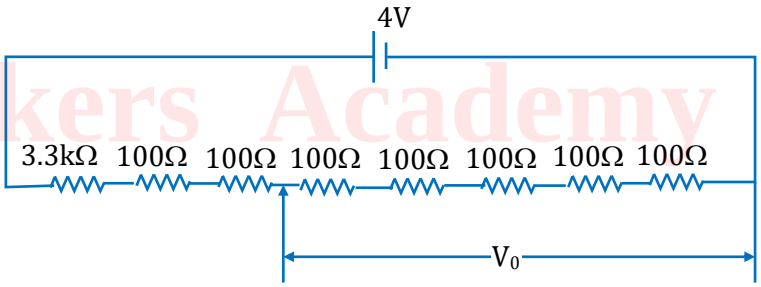
(A) $V_1 = V_2$ (B) $V_1 \neq V_3 - V_2$ (C) $V_1 + V_2 > V_3$ (D) $V_1 + V_2 = V_3$

PCT160

2. Two resistance of 100 Ω and 200 Ω are connected in series with a battery of 4 V and negligible internal resistance. A voltmeter is used to measure voltage across 100 Ω resistance, which gives reading as 1 V. The resistance of voltmeter must be_____Ω.

PCT161

3. A potential divider circuit is shown in figure. The output voltage V_0 is

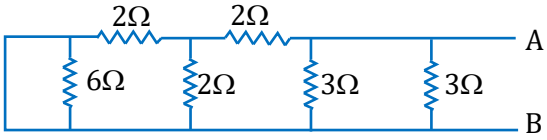


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(A) 4V (B) 2 mV (C) 0.5 V (D) 12 mV

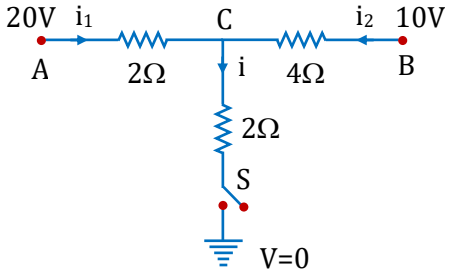
PCT162

4. Equivalent resistance of the following network is ____ Ω.



PCT163

5. When the switch S, in the circuit shown, is closed, then the value of current i will be :

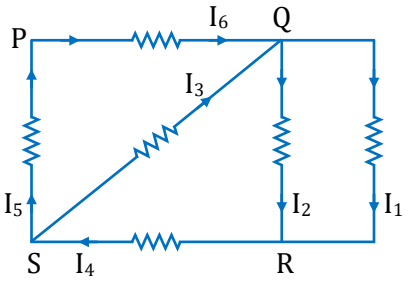


(A) 3 A (B) 5 A (C) 4 A (D) 2 A

PCT164

For More Join: @IITJEE_Advanced

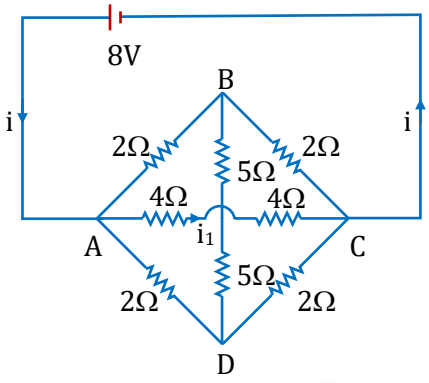
6. In the given circuit diagram, the currents, $I_1 = -0.3A$, $I_4 = 0.8\ A$ and $I_5 = 0.4\ A$, are flowing as shown. The currents I_2, I_3 and I_6, respectively, are:



(A) 1.1 A, 0.4 A, 0.4 A (B) –0.4 A, 0.4 A, 1.1 A (C) 0.4 A, 1.1 A, 0.4 A (D) 1.1 A, –0.4 A, 0.4 A

PCT165

7. The value of current i_1 flowing from A to C in the circuit diagram is :

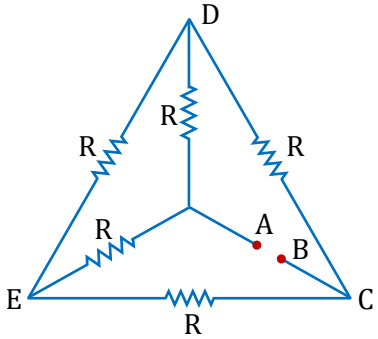


(A) 5A (B) 2A (C) 4A (D) 1A

PCT166

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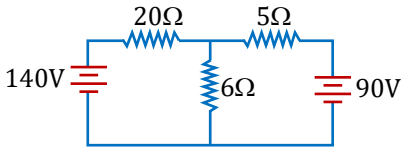
8. Five equal resistances are connected in a network as shown in figure. The net resistance between the points A and B is :



(A) $2R$ (B) $\frac{R}{2}$ (C) $\frac{3R}{2}$ (D) R

PCT167

9. The value of current in the 6Ω resistance is :

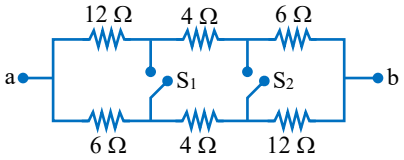


(A) 4A (B) 8A (C) 10A (D) 6A

PCT168

For More Join: @IITJEE_Advanced

10. In the given figure switches S_1 and S_2 are in open condition. The resistance across ab when the switches S_1 and S_2 are closed is _____Ω.



PCT169

11. If you are provided a set of resistances 2Ω, 4Ω, 6Ω and 8Ω. Connect these resistances so as to obtain an equivalent resistance of $\frac{46}{3}\ \Omega$.

(A) 4Ω and 6Ω are in parallel with 2Ω and 8 Ω in series

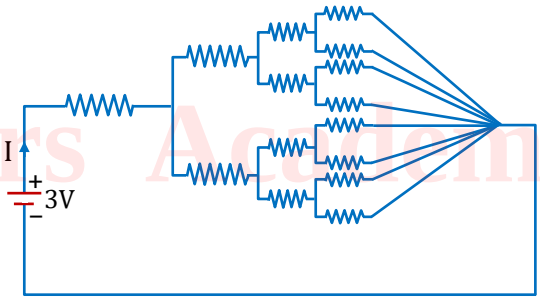
(B) 6Ω and 8Ω are in parallel with 2Ω and 4Ω in series

(C) 2Ω and 6Ω are in parallel with 4Ω and 8 Ω in series

(D) 2Ω and 4Ω are in parallel with 6Ω and 8Ω in series

PCT170

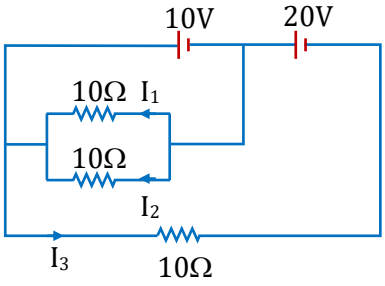
12. All resistances in figure are 1Ω each. The value of current 'I' is $\frac{a}{5}A$. The value of a is _____



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PCT171

13. In the given circuit the value of $\left|\frac{I_1+I_3}{I_2}\right|$ is:



PCT172

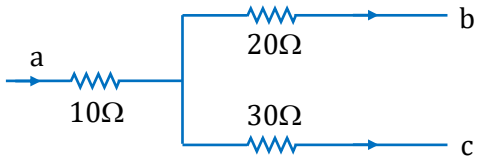
14. A cell of emf 90 V is connected across series combination of two resistors each of 100Ω resistance. A voltmeter of resistance 400 Ω is used to measure the potential difference across each resistor. The reading of the voltmeter will be :

(A) 40 V (B) 45 V (C) 80 V (D) 90 V

PCT173

For More Join: @IITJEE_Advanced

15. Figure shows a part of an electric circuit. The potentials at points a, b and c are 30 V, 12 V and 2V respectively. The current through the 20Ω resistor will be.



(A) 0.4 A (B) 0.2 A (C) 0.6 A (D) 1.0 A

PCT174

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ANSWER KEY

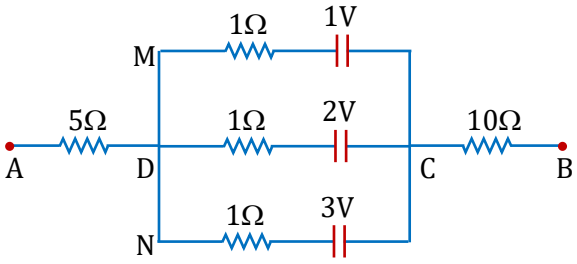
Question	1	2	3	4	5	6	7	8	9	10
Answer	D	200	C	1	B	A	D	D	C	10
Question	11	12	13	14	15					
Answer	D	8	2	A	A					

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Current Electricity : RACE-03

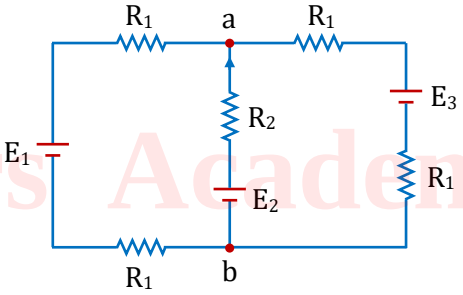
1. In the circuit, the potential difference between A and B is :-



(A) 6 V (B) 1 V (C) 3 V (D) 2 V

PCT175

2. For the circuit shown, with $R_1 = 1.0\Omega$, $R_2 = 2.0\ \Omega$, $E_1 = 2\ V$ and $E_2 = E_3 = 4\ V$, the potential difference between the points 'a' and 'b' is approximately (in V) :



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(A) 2.7 (B) 3.3 (C) 2.3 (D) 3.7

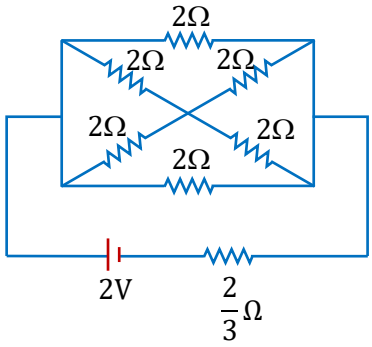
PCT176

3. By what percentage will the illumination of the lamp decrease if the current drops by 20%?

(A) 46% (B) 26% (C) 36% (D) 56%

PCT177

4. In the following circuit, the battery has an emf of 2 V and an internal resistance of $\frac{2}{3}\Omega$. The power consumption in the entire circuit is _____ W.



PCT178

For More Join: @IITJEE_Advanced

5. A heating element has a resistance of 100Ω at room temperature. When it is connected to a supply of 220 V, a steady current of 2A passes in it and temperature is 500ºC more than room temperature. What is the temperature coefficient of resistance of the heating element :-

(A) $1\times10^{-4}\,°C^{-1}$ (B) $2\times10^{-4}\,°C^{-1}$ (C) $0.5\times10^{-4}\,°C^{-1}$ (D) $5\times10^{-4}\,°C^{-1}$

PCT179

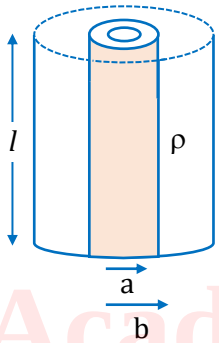
6. Two electric bulbs, rated at (25 W, 220 V) and (100 W, 220 V), are connected in series across a 220 V voltage source. If the 25 W and 100 W bulbs draw powers P_1 and P_2 respectively, then:

(A) $P_1 = 9\ W, P_2 = 16\ W$ (B) $P_1 = 4\ W, P_2 = 16W$

(C) $P_1 = 16\ W, P_2 = 4W$ (D) $P_1\ 16\ W, P_2 = 9W$

PCT180

7. Model a torch battery of length l to be made up of a thin cylindrical bar of radius 'a' and a concentric thin cylindrical shell of radius 'b' filled in between with an electrolyte of resistivity ρ (see figure). If the battery is connected to a resistance of value R, the maximum Joule heating in R will take place for:-



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(A) $R = \frac{2\rho}{\pi l} ln\left(\frac{b}{a}\right)$ (B) $R = \frac{\rho}{\pi l} ln\left(\frac{b}{a}\right)$ (C) $R = \frac{\rho}{2\pi l}\left(\frac{b}{a}\right)$ (D) $R = \frac{\rho}{2\pi l} ln\left(\frac{b}{a}\right)$

PCT181

8. A uniform heating wire of resistance 36Ω is connected across a potential difference of 240 V. The wire is then cut into half and potential difference of 240 V is applied across each half separately. The ratio of power dissipation in first case to the total power dissipation in the second case would be 1 : x, where x is _____.

PCT182

9. Given below are two statements:

Statement I : A uniform wire of resistance 80Ω is cut into four equal parts. These parts are now connected in parallel. The equivalent resistance of the combination will be 5Ω.

Statement II : Two resistance 2R and 3R are connected in parallel in a electric circuit. The value of thermal energy developed in 3R and 2R will be in the ratio 3 : 2.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both statement I and statement II are correct.

(B) Both statement I and statement II are incorrect.

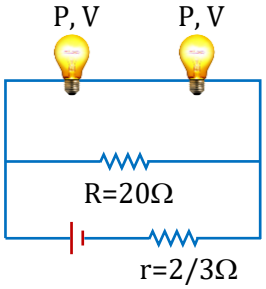
(C) Statement I is correct but statement II is incorrect.

(D) Statement I is incorrect but statement II is correct.

PCT183

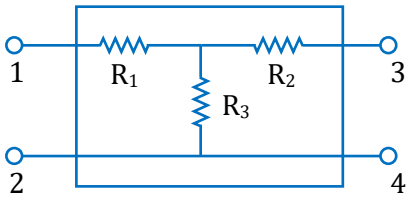
For More Join: @IITJEE_Advanced

10. Two bulbs, each of rating 10 V, 50 W, are connected as shown. To make the bulbs glow at their full intensity, find the emf (in Volt) of the battery connected.



PCT184

11. A concealed circuit (a black box) consisting of resistors has four terminals. If a voltage is applied between clamps 1 and 2 when clamps 3 and 4 are open, the power liberated is $P_1 = 40\ W$ and when clamps 3 and 4 are closed, the power liberated is $P_2 = 80W$. If the same source is connected to the clamps 3 and 4, the power liberated in the circuit when clamps 1 and 2 are open is $P_3 = 20W$. Determine the power P_4 (in Watt) consumed in the circuit when the clamps 1 and 2 are connected and the same voltage is applied between the clamps 3 and 4.



PCT185

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12. An enquiring physics student connects a cell to a circuit and measures the current drawn from the cell to I_1. When he joins a second identical cell is series with the first, the current becomes I_2. When the cells are connected are in parallel, the current through the circuit is I_3. Relation between the current is $A\ I_3\ I_2 = B\ I_1\ (I_2 + I_3)$. Find A/B.

PCT186

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	B	C	3	B	C	D	4	C	24
Question	11	12								
Answer	40	1.5								

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Current Electricity : RACE-04

1. In a metre-bridge when a resistance in the left gap is 2Ω and unknown resistance in the right gap, the balance length is found to be 40 *cm*. On shunting the unknown resistance with 2Ω, the balance length changes by:

(A) 22.5 *cm* (B) 20 *cm* (C) 62.5 *cm* (D) 65 *cm*

PCT187

2. A current of 200 μ*A* deflects the coil of a moving coil galvanometer through 60^0. The current to cause deflection through $\frac{\pi}{10}$ radian is:

(A) 30 μ*A* (B) 120 μ*A* (C) 60 μ*A* (D) 180 μ*A*

PCT188

3. The deflection in moving coil galvanometer falls from 25 divisions to 5 division when a shunt of 24Ω is applied. The resistance of galvanometer coil will be :

(A) 12Ω (B) 96Ω (C) 48Ω (D) 100Ω

PCT189

4. When a current of 5 *mA* is passed through a galvanometer having a coil of resistance 15 Ω, it shows full scale deflection. The value of the resistance to be put in series with the galvanometer to convert it into to voltmeter of range 0-10 *V* is:

(A) $2.535 \times 10^3\ \Omega$ (B) $4.005 \times 10^3\ \Omega$ (C) $1.985 \times 10^3\ \Omega$ (D) $2.045 \times 10^3\ \Omega$

PCT190

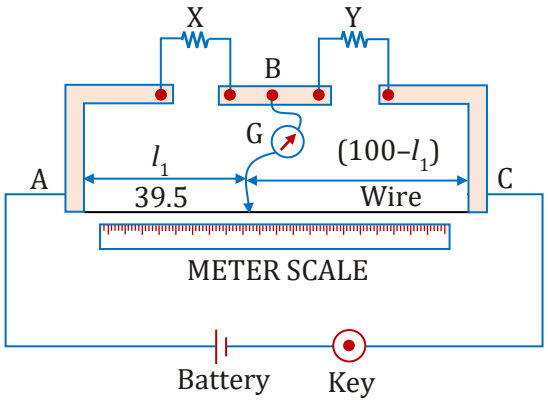
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5. On interchanging the resistances, the balance point of a meter bridge shifts to the left by 10 *cm*. The resistance of their series combination is 1 *k*Ω. How much was the resistance on the left slot before interchanging the resistances ?

(A) 505 *k*Ω (B) 550 *k*Ω (C) 910 *k*Ω (D) 990 *k*Ω

PCT191

6. In a meter bridge, as shown in the figure, it is given that resistance Y = 12.5 Ω and that the balance is obtained at a distance 39.5 *cm* from end A (by Jockey J). After interchanging the resistances X and Y, a new balance point is found at a distance l_2 from end A. What are the values of X and l_2?

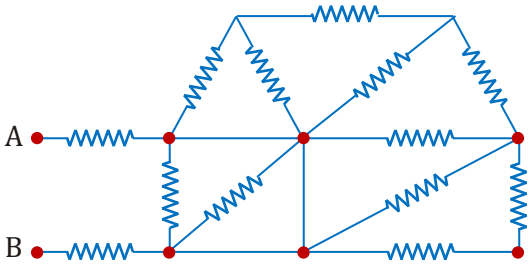


(A) 19.15 Ω and 60.5 *cm* (B) 8.16 Ω and 60.5 *cm*
(C) 8.16 Ω and 39.5 *cm* (D) 19.15 Ω and 39.5 *cm*

PCT192

For More Join: @IITJEE_Advanced

7. In the given circuit all resistances are of value R ohm each. The equivalent resistance between A and B is :



(A) $2R$ (B) $\frac{5R}{2}$ (C) $3R$ (D) $\frac{5R}{3}$

PCT193

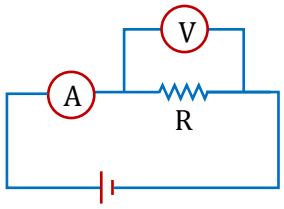
8. In a circuit for finding the resistance of a galvanometer by half deflection method, a 6 V battery and a high resistance of 11 $k\Omega$ are used. The figure of merit of the galvanometer is 60μA/division. In the absence of shunt resistance, the galvanometer produces a deflection of $\theta = 9$ divisions when current flows in the circuit. The value of the shunt resistance that can cause the deflection of $\theta/2$, is closest to :-

(A) 55 Ω (B) 110 Ω (C) 220 Ω (D) 550 Ω

PCT194

9. The actual value of resistance R, shown in the figure is 30Ω. This is measured in an experiment as shown using the standard formula $R = \frac{V}{I}$, where V and I are the readings of the voltmeter and ammeter, respectively. If the measured value of R is 5% less, then the internal resistance of the voltmeter is:

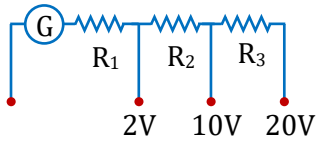
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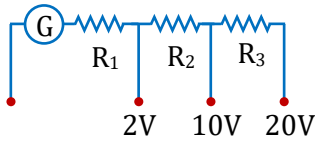
(A) 350Ω (B) 570Ω (C) 35Ω (D) 600Ω

PCT195

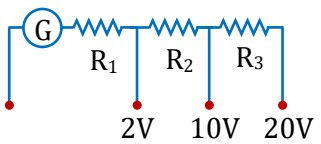
10. A galvanometer of resistance 100Ω has 50 divisions on its scale and has sensitivity of 20 μA/division. It is to be converted to a voltmeter with three ranges, of 0–2 V, 0–10 V and 0–20 V. The appropriate circuit to do so is :

(A) 

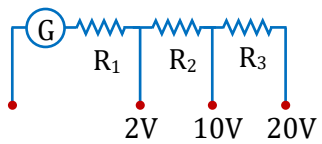
R_1=1900Ω
R_2=9900Ω
R_3=19000Ω

(B) 

R_1=2000Ω
R_2=8000Ω
R_3=10000Ω

(C) 

R_1=19900Ω
R_2=9900Ω
R_3=19000Ω

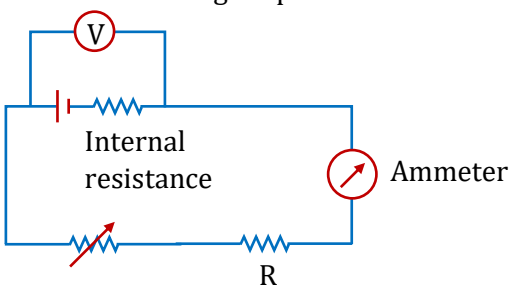
(D) 

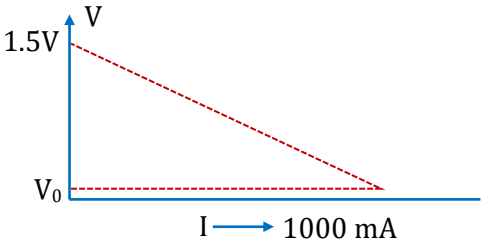
R_1=1900Ω
R_2=8000Ω
R_3=10000Ω

PCT196

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11. To verify Ohm's law, a student connects the voltmeter across the battery as, shown in the figure. The measured voltage is plotted as a function of the current, and the following graph is obtained:





If V_0 is almost zero, identify the correct statement:

(A) The value of the resistance R is 1.5 Ω.

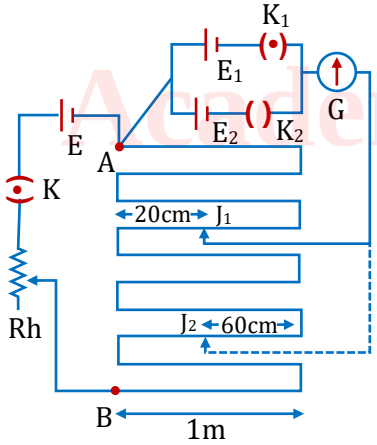
(B) The emf of the battery is 1.5 V and the value of R is 1.5 Ω.

(C) The emf of the battery is 1.5 V and its internal resistance is 1.5 Ω.

(D) The potential difference across the battery is 1.5 V when it sends a current of 1000 mA.

PCT197

12. In the given circuit of potentiometer, the potential difference E across AB (10m length) is larger than E_1 and E_2 as well. For key K_1 (closed), the jockey is adjusted to touch the wire at point J_1 so that there is no deflection in the galvanometer. Now the first battery (E_1) is replaced by second battery (E_2) for working by making K_1 open and K_2 closed. The galvanometer gives then null deflection at J_2. The value of $\frac{E_1}{E_2}$ is $\frac{a}{b}$, where a = _____.



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PCT198

13. A teacher in his physics laboratory allotted an experiment to determine the resistance (G) of a galvanometer. Students took the observations for $\frac{1}{3}$ deflection in the galvanometer. Which of the below is **true** for measuring value of G ?

(A) $\frac{1}{3}$ deflection method cannot be used for determining the resistance of the galvanometer.

(B) $\frac{1}{3}$ deflection method can be used and in this case the G equals to twice the value of shunt resistance(s).

(C) $\frac{1}{3}$ deflection method can be used and in this case, the G equals to three times the value of shunt resistance(s).

(D) $\frac{1}{3}$ deflection method can be used and in this case the G value equals to the shunt resistance(s).

PCT199

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14. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : For measuring the potential difference across a resistance of 600 Ω, the voltmeter with resistance 1000 Ω will be preferred over voltmeter with resistance 4000 Ω.

Reason R : Voltmeter with higher resistance will draw smaller current than voltmeter with lower resistance.

In the light of the above statements, choose the **most appropriate** answer from the options given below.

(A) **A** is not correct but **R** is correct.

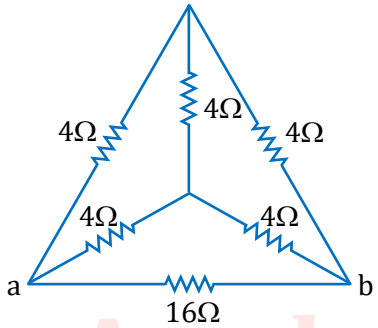
(B) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

(C) Both **A** and **R** are correct but **R** is not the correct explanation of **A**.

(D) **A** is correct but **R** is not correct.

PCT200

15. The equivalent resistance of the circuit shown below between points *a* and *b* is :



(A) 24Ω (B) 3.2Ω (C) 20Ω (D) 16Ω

PCT201

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	B	C	B	B	A	B	B	D
Question	11	12	13	14	15					
Answer	C	1	B	A	B					

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